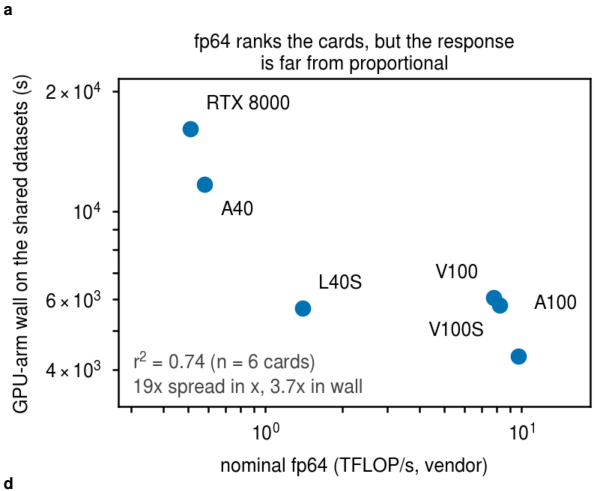
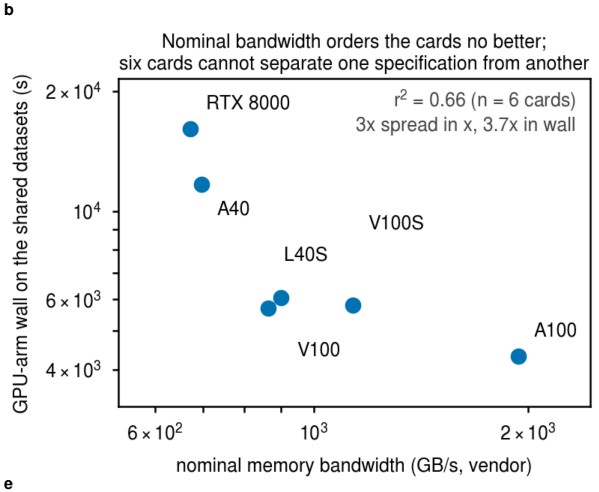
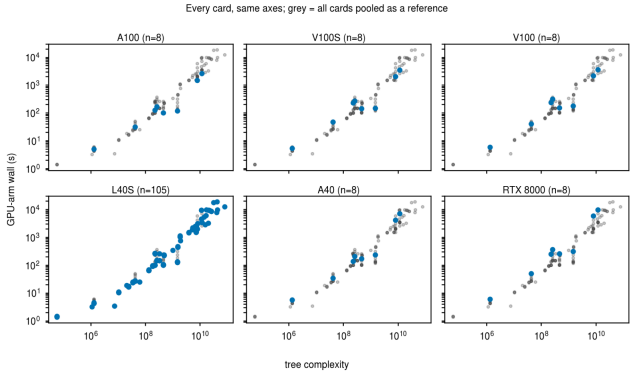
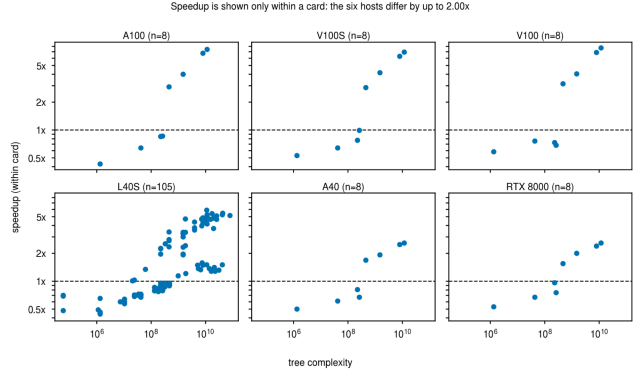
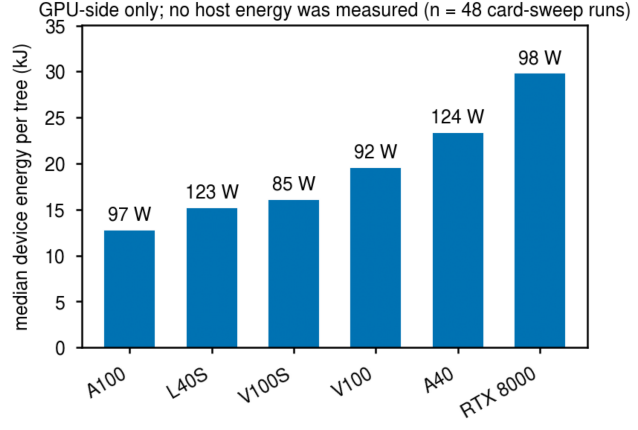
Figure 5


a
fp64 ranks the cards, but the response
is far from proportional
RTX 8000
A40
L40S
V100
A100
V100S
r² = 0.74 (n = 6 cards)
19x spread in x, 3.7x in wall
GPU-arm wall on the shared datasets (s)
nominal fp64 (TFLOP/s, vendor)


b
Nominal bandwidth orders the cards no better;
six cards cannot separate one specification from another
RTX 8000
A40
L40S
V100S
V100
A100
r² = 0.66 (n = 6 cards)
3x spread in x, 3.7x in wall
GPU-arm wall on the shared datasets (s)
nominal memory bandwidth (GB/s, vendor)


c
Every card, same axes; grey = all cards pooled as a reference
A100 (n=8)
V100S (n=8)
V100 (n=8)
L40S (n=105)
A40 (n=8)
RTX 8000 (n=8)
GPU-arm wall (s)
tree complexity


d
Speedup is shown only within a card: the six hosts differ by up to 2.00x
A100 (n=8)
V100S (n=8)
V100 (n=8)
L40S (n=105)
A40 (n=8)
RTX 8000 (n=8)
speedup (within card)
tree complexity


e
Energy to produce the same trees, by card
GPU-side only; no host energy was measured (n = 48 card-sweep runs)
97 W
123 W
85 W
92 W
124 W
98 W
median device energy per tree (kJ)
A100
L40S
V100S
V100
A40
RTX 8000